

Kumbha Nagaswetha

Data Scientist at Atlas Copco Group
M.Tech from Indian Institute of Science, Bangalore
Specialization in Computer Vision, Deep Learning

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LinkedIn Profile

Education

- Indian Institute of Science, Bengaluru** 2020-2022
M.Tech in Signal Processing - CGPA : 7.0 Bengaluru, KA
- Jawaharlal Nehru Technological University Kakinada** 2016-2020
B.Tech in Electronics and Communication Engineering - CGPA : 8.27 Kakinada, AP

Experience

- Atlas Copco (India), Pune** 2022 - Present
Data Scientist with 3 years of experience, specializing in Machine Learning, Computer Vision, and Optimization Techniques. Skilled in developing end-to-end ML pipelines for real-world applications, including machine health monitoring, survival analysis, surrogate modeling of physical processes. Proficient in Deep Learning (NNs, CNNs, GANs), Signal Processing, Image Processing, Anomaly Detection, and Domain Adaptation, with expertise in leveraging techniques like t-SNE and SHAP for model interpretability. Also in process of filing a patent on algorithm for AI based Design Optimization.

Projects (Atlas Copco (India))

- Machine Health Monitoring Using Domain Adaptation** June.2024–Present
 - Developing a domain adaptation model to identify valve damages in piston compressor across varying operating conditions, eliminating the need for multiple models and reducing distribution difference between different datasets, using domain transfer analysis on spectrograms of Vibration data.
 - Explored anomaly detection and GAN-based techniques to improve model generalization across domains. Employed t-SNE, LIME and SHAP to interpret CNN-based model decisions and gain insights into learned features.
 - Utilizing limited training data to achieve 80% accuracy in damage detection.
- AI based Design Optimization** November.2024–Present
 - Implementing the Bayesian method with multi-surrogate method for shape optimization of Jet Pump to find the best shape with least number of function evaluation and parallel computation in HPC cluster.
 - Utilized the Constant Liar approach with Kriging, RBNN, Random Forest, and SVR surrogates, achieving the best efficiency with minimal iterations.
 - Best shape found within 30-40 iterations.
 - Explored, compared and documented pros and cons of sampling methods (LHS, CVT, McQueen, etc.), surrogate models (Kriging, Radial Basis Functions, NNs, etc.) and optimization techniques (Bayesian, GA, PSO, BFGS) for optimization problems.
- Speed Estimation of a Compressor from Vibration Data using ML** Sep.2022–March.2023
 - Developed a pipeline to transform vibration signal frequency spectra, highlighting the compressor's speed and harmonics.
 - Designed a custom CNN for feature extraction and integrated it with LightGBM for classification.
 - Achieved 99.5% classification accuracy, with only 3 misclassified samples out of 600.
- Lubricant Level Estimation Using CNN on Mel-spectrogram of Audio Signals** March.2023–Sep.2023
 - Led the development and implementation of the CNN architecture for Mel spectrogram analysis.
 - Utilized Python, TensorFlow, and Keras for model training; employed Librosa for audio processing and VGG16 for feature extraction.
 - Achieved an R^2 score of 0.92 on the test set; deployed a pyfunc-based pipeline with a user-friendly interface.
- Survival Analysis for Predictive Maintenance** April.2023– July.2023
 - Performed survival analysis to predict the lifecycle of critical components, improving maintenance planning and resource allocation.
 - Leveraged the `randomForestSRC` package in R for time-to-event predictions.
 - Developed an interactive Shiny web application for real-time data entry and visualization of survival analysis results.
- Sales Forecasting and Demand Planning** August.2023–November.2023
 - Built a data-driven forecasting model to optimize sales strategy and decision-making.
 - Identified leading indicators for opportunity-to-sales conversion, achieving 87% accuracy using a Random Forest classifier.
 - Forecasted material orders using SARIMAX, ARIMA, LSTM, NN, n-hits and TCN models, achieving 90% accuracy with external indicators like inflation rate and GDP.

Projects (Academics)

- **HDR Image Reconstruction using Un-supervised Learning** IISc, Dec 2021 - June 2022
 - HDR image is reconstructed from two different exposure time images using Unsupervised Learning. Created a model similar to VGG 16 and optimized a customized loss function within the model. The important features are reconstructed and fusion is done to get the HDR image.
- **Facial Emotion Recognition** IISc, Feb 2021 - Jun 2021
 - Classifies the face emotion based on the facial expression. Worked on three datasets JAFFE, AFED, and FER. and used Fine-tuning on Alex Net and SVM for classification.
- **A Fuzzy Based DWM filter For Impulse Noise Removal** JNTUK, Dec 2019 - Mar 2020
 - Impulse noise removal is removed using a Fuzzy based Directional Weighted Median Filter. First the image is divided into smooth and non-smooth regions based on standard deviation. Then the detected noise is removed by a directional weighted median filter for non-smooth region and by a median filter for a smooth region.
- **Other Mini-projects**
 - Features detection, Image segmentation, Object detection, JPEG implementation, HDR imaging techniques, Image noise removal, Classification, Regressions, clustering , Binary morphology, Image filtering.

Technical Skills

Technology: Machine Learning, Deep Learning, Computer Vision, Image Processing, Signal Processing, HPC, Databricks

Languages: Python, C, MATLAB, PySpark

Libraries: OpenCV, Tensorflow, Pytorch, scikit-learn, scipy, imblearn, Pyspark, Pandas, NumPy etc.

Major Courses Undertaken: Digital Image processing, Advanced Image Processing, Computer Vision, Pattern Recognition and Neural Networks.

Extra-Curricular Activities

- Participated in internship at BSNL Telecommunications in Kakinada.
- Co-ordinator of the technical event related to the department of ECE in the Annual techno-cultural fest, JNTU Kakinada, 2020.
- Attained yellow grade in Taekwondo federation of India.
- Participated in department level chess competitions.